Experiment 2: 8 Queen Problem (Python)

Aim

To write a Python program to solve the **8 Queen Problem** using **Backtracking**.

Algorithm

1. Start placing queens from the first row.
2. Check if the current position is safe.
3. If safe, place the queen.
4. Move to the next row.
5. If no safe position is found, backtrack and try another position.
6. Repeat until all 8 queens are placed.
7. Print the solution.

PROGRAM

```
N = 8


def safe(board, row, col):
    for i in range(row):
        if board[i] == col or abs(board[i] - col) == abs(row - i):
            return False
    return True


def solve(board, row):
    if row == N:
        for i in range(N):
            for j in range(N):
                if board[i] == j:
                    print("Q", end=" ")
                else:
                    print(".", end=" ")
```

```
        print()
      return True

    for col in range(N):
      if safe(board, row, col):
        board[row] = col
        if solve(board, row + 1):
          return True
        board[row] = -1
    return False

board = [-1] * N
solve(board, 0)

board = [-1] * N
solve(board, 0)
```

INPUT

NO input required

OUTPUT

```
Q . . . . . . .
. . . . Q . . .
. . . . . . . Q
. . . . . Q . .
. . Q . . . . .
. . . . . . Q .
. Q . . . . . .
. . . Q . . . .
```